Hershey Beatrice Marbeda

CAPSTONE LAB 3

This capstone lab covers topics learned in Chapters 9 through 12 of *Guide to TCP/IP: IPv6 and IPv4*. It assumes you are working in a Windows 7 Professional or Windows 10 Pro environment. See the Preface for details about system requirements. To complete this lab, you must have Wireshark for Windows software installed and the trace (data) file named capstone-labs-3-trace-file.pcapng.

Capstone Lab 3.1

Capstone Lab 3.1: Configure Wireshark Display Filters to View TCP/IP Transport Layer Protocols

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view specific applications that use TCP/IP Transport layer protocols.

Description: In this lab, you will create several display filters in Wireshark and apply them to a trace file to view specific applications using TCP/IP Transport layer protocols (User Datagram Protocol [UDP] or Transmission Control Protocol [TCP]). You need the capstone-labs-3-trace-file.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-3-trace-file.pcapng** file.
2. Configure individual display filters to find the following application packets:
    1. Dynamic Host Configuration Protocol (DHCP)
    2. Dynamic Host Configuration Protocol version 6 (DHCPv6)
    3. Domain Name System (DNS) queries
    4. Hypertext Transfer Protocol (HTTP)
    5. Simple Network Management Protocol (SNMP)
    6. Telnet

7. Trivial File Transfer Protocol (TFTP)

3. Apply one of the filters to the trace file.
4. Browse the list of application packets and select one. (The packet to select should have a source address of 0.0.0.0 for DHCP and fe80::958c:8d52:57d:fe80 for DHCPv6. For all other applications, the source address is either 10.1.101.100 or 2001:db8:1ab:1001::108If both addresses are available for an application, look at each one individually.)
5. In the following table, record the following information for the application type:
    1. Is this application using UDP or TCP?
    2. Is this application using IPv4, IPv6, or both?
    3. What is the destination port for this application?
6. Clear the display filter.
7. Repeat steps 3 through 6 for each filter/application type.
8. What did you accomplish in this part of Capstone Lab 3? Summarize your answer.
9. Clear the display filter box.

Question 5/7 – Please fill out the table. You will be repeating steps 3-6 to complete this.

Application	UDP or TCP	IPv4, IPv6, or Both?	Destination Port
DHCP	UDP	IPv4	67
DHCPv6	UDP	IPv6	547
DNS	UDP	BOTH	53
HTTP	TCP	BOTH	80
SNMP	UDP	IPv4	161
Telnet	TCP	BOTH	23
TFTP	UDP	BOTH	69

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Question 8 – Please answer lab summary question.

I analyzed different network protocols to determine if they are UDP or TCP or BOTH, IP versions, and destination ports.

Capstone Lab 3.2

Capstone Lab 3.2: Create a Wireshark Display Filter to View DHCPv6 Messages

Time Required: 15 minutes

Objective: View a screen capture output from a Windows 10 host, and then use Wireshark to view a trace file and configure a display filter to view DHCPv6 messages from the same host.

Description: In this lab, you will view Ipconfig screen capture output from a Windows 10 host and then use Wireshark to configure a specific display filter to view DHCPv6 messages. During the lab, you will discover when the Windows 10 host was specifically assigned its IPv6 address as seen in the Ipconfig output. In the trace file, you may notice that this host has been assigned different IPv6 addresses at different times. You need the capstone-labs-3-trace-file.pcapng file to complete the lab.

1. Review the Ipconfig output in Figure CL3-1 and record the IPv6 Global Unicast Address (GUA) that is assigned to this Windows 10 host.

```
Connection-specific DNS Suffix  . : ipv6sandbox.com
Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Network Connection
Physical Address. . . . . . . . . : 00-0C-29-3C-19-C7
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . . . . . . . : 2001:db8:1ab:1001::108(Preferred)
Lease Obtained. . . . . . . . . . : Wednesday, June 08, 2016 14:35:46
Lease Expires . . . . . . . . . . : Wednesday, June 08, 2016 15:56:46
Link-local IPv6 Address . . . . . : fe80::958c:8d52:57d:fe80%6(Preferred)
IPv4 Address. . . . . . . . . . . : 10.1.101.100(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Lease Obtained. . . . . . . . . . : Wednesday, June 08, 2016 14:35:30
Lease Expires . . . . . . . . . . : Wednesday, June 08, 2016 16:20:30
Default Gateway . . . . . . . . . : fe80::f252%6
                                    10.1.101.1
DHCP Server . . . . . . . . . . . : 10.1.0.200
DHCPv6 IAID . . . . . . . . . . . : 234884137
DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-1E-C1-D4-9C-00-0C-29-3C-19-C7
DNS Servers . . . . . . . . . . . : 2001:db8:1ab:ba5e::2000
                                    10.1.0.200
NetBIOS over Tcpip. . . . . . . . : Enabled
Connection-specific DNS Suffix Search List :
                                    ipv6sandbox.com

C:\>
```

Figure CL3-1 Ipconfig output

```
IPv6 Address. . . . . . . . . . . : 2001:db8:1ab:1001::108(Preferred)
```

The IP we analyzed

Figure CL3-1 Ipconfig outputIn Wireshark, open the **capstone-labs-3-trace-file.pcapng** file, if it's not already open.

2. Configure a display filter to find all DHCPv6 packets.
3. Apply the filter to the trace file.



4. With the DHCPv6 packets displayed, what packet is the first one where the DHCPv6 server supplies the Windows 10 host with its address? You will need to look for DHCPv6 server Reply messages, expand **DHCPv6** in the packet details

pane, expand **Identity Association for Non-temporary Address**, and expand **IA Address** in order to view the assigned IPv6 address.

5. Bonus question: What specific display filter could you create to find the answer in Step 5 easier or quicker?
6. What did you accomplish in this part of Capstone Lab 3? Summarize your answer.
7. Clear the display filter box.

Question 5 – Please answer the question.

Frame 9294

(Optional Bonus Question) Question 6 – Please answer the question.



Question 7 – Please answer lab summary question.

I learned how to filter out DHCPv6 and to look for an associated IPv6 address, which is 2001:db8:1ab:1001::10

Capstone Lab 3.3

Capstone Lab 3.3: Create a Wireshark Display Filter to View Router Advertisement Messages

Time Required: 15 minutes

Objective: View a screen capture output from a Layer 3 switch (router), and then use Wireshark to view a trace file and configure a display filter to view Router Advertisement messages from the router.

Description: In this lab, you will view configuration screen capture output from a Layer 3 switch (router). You will then use Wireshark to configure a specific display filter to view Router Advertisement messages, and then verify that the router configuration as

seen is producing the Router Advertisement messages as viewed. You need the capstone-labs-3-trace-file.pcapng file to complete the lab.

1. Review the Layer 3 switch configuration screen capture in Figure CL3-2. Looking in the "vlan 2" portion of the configuration, what are the M, O, L, and A flag settings? Record these values. You may notice that you don't see configuration statements or definitions for the O and L flags. What do you suspect they are set to?

```
web-management ssl
ip dns server-address priority 1 2001:db8:1ab:ba5e::2000
ip dns server-address priority 2 10.1.0.200
ip router-id 10.0.0.252
ip routing
ipv6 unicast-routing
snmp-server community "public" unrestricted
router ospf
   area 0.0.0.2
   area backbone
   enable
   exit
router ospf3
   area 0.0.0.2
   area backbone
   enable
   exit
vlan 1
   name "DEFAULT_VLAN"
   no untagged 1-6,12
   untagged 7-11,13-24
   ip address 10.0.0.252 255.255.255.0
   ip ospf 10.0.0.252 area backbone
   ipv6 enable
   ipv6 address fe80::f252 link-local
   ipv6 address 2001:db8:1ab:c07e::f252/64
   ipv6 ospf3 area backbone
   ipv6 nd ra suppress
   exit
vlan 2
   name "lab-1001"
   untagged 1-6
   ip address 10.1.101.1 255.255.255.0
   ip helper-address 10.1.0.200
   ip ospf 10.1.101.1 area 0.0.0.2
   ipv6 enable
   ipv6 address fe80::f252 link-local
   ipv6 address 2001:db8:1ab:1001::f252/64
   ipv6 ospf3 area 0.0.0.2
   ipv6 nd ra managed-config-flag
   ipv6 nd ra max-interval 200
   ipv6 nd ra min-interval 110
   ipv6 nd ra prefix 2001:db8:1ab:1001::/64 40 20 no-autoconfig
   ipv6 helper-address unicast 2001:db8:1ab:ba5e::2000
   exit
dhcpv6-relay
```

Figure CL3-2 Layer 3 switch configurationIn Wireshark, open the **capstone-labs-3-trace-file.pcapng** file, if it's not already open.

2. Configure a display filter to find ICMPv6 type 134 Router Advertisement packets.
3. Apply the filter to the trace file.

4. Select one of the Router Advertisement packets and observe its structure. In the packet details pane, expand **Internet Control Message Protocol v6** and then expand **Flags** to display the settings for the Managed address configuration and Other configuration options. From there, expand **ICMPv6 Option (Prefix information)** and then expand **Flag** to display the settings for the On-link flag(L) and Autonomous address-configuration flag(A) options.
5. While viewing this Router Advertisement packet, record the settings of the M, O, L, and A flags. Compare the settings in the Router Advertisement packet to the settings you recorded from the Layer 3 switch configuration in Step 1.
6. Do you see anything different in the settings between these two views?
7. Bonus question: Is this Layer 3 switch providing the DHCPv6 services for hosts in vlan 2 or is it providing DHCPv6 relay functions? If you believe that it is providing DHCPv6 relay functions, what DHCPv6 server does it send the hosts DHCPv6 requests to?
8. What did you accomplish in this part of Capstone Lab 3? Summarize your answer.
9. Clear the display filter box.

Question 1a – Please answer the question, including which flags, and what they do.

M, O, L, A

M = 1, O = ?, L = ?, A = 0

```
ipv6 nd ra managed-config-flag
ipv6 nd ra max-interval 200
ipv6 nd ra min-interval 110
ipv6 nd ra prefix 2001:db8:1ab:1001::/64 40 20 no-autoconfig
ipv6 helper-address unicast 2001:db8:1ab:ba5e::2000
```

Question 1b – Please answer the question about the O and L flags.

M = 1, O = 0, L = ?, A = 0

```
    1... .... = Managed address configuration: Set
    .0.. .... = Other configuration: Not set
```

Question 6 - Please answer the question.

M = 1, O = 0, L = ?, A = 0

```
Flag: 0x80, On-link Flag (L)
  1... .... = On-link Flag (L): Set
```

Therefore, L = 1

Question 7 - Please answer the question.

M and A are exactly the same. L is also ON as we expected, and O was OFF as expected.

(Optional Bonus Question) Question 8 - Please answer the question.

```
ipv6 helper-address unicast 2001:db8:1ab:ba5e::2000
```

It is a DHCP relay as it is shown above

Question 9 (Required) – Please answer lab summary question.

I learned how to recognize and analyze the flag messages under ICMPv6 == 134 or the Router Advertisement messages

Capstone Lab 3.4

Capstone Lab 3.4: Configure Wireshark Display Filters to View Secure TCP/IP Transport Layer Protocol

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view specific secure applications using TCP/IP Transport Layer Protocol.

Description: When accessing infrastructure devices inside your network, it is considered a best practice to do so using secure applications such as ones that support

Secure Shell (SSH) and Hyper Text Transfer Protocol Secure (HTTPS). In this lab, you will create display filters in Wireshark and apply them to a trace file to view specific secure applications using Transmission Control Protocol (TCP), a TCP/IP Transport layer protocol. You need the capstone-labs-3-trace-file.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-3-trace-file.pcapng** file, if it's not already open.
2. Configure a display filter to find all SSH packets.
3. Apply the filter to the trace file.
4. View the list of SSH packets and select the first packet that has an IPv6 source address of 2001:db8:1ab:1001::108. What is this packet number?
5. Looking in the Bytes pane, select each of the 10 subsequent packets. Can you see any data/information in clear text, such as login name, devices information, and so on?
6. Scroll through the list of ssh packets and select the first packet that has an IPv4 source address of 10.1.101.100. What is this packet number?
7. Looking in the Bytes pane, select each of the 10 subsequent packets. Can you see any data/information in clear text, such as login name, devices information, and so on?
8. Clear the display filter box.
9. Configure a display filter to find all HTTPS packets. (Hint: Wireshark does not have a display for HTTPS. You must configure a specific TCP port 443 display filter.)
10. Apply the filter to the trace file.
11. View the list of HTTPS packets and select the first packet that has an IPv4 source address of 10.1.101.100. What is this packet number?
12. Looking in the Bytes pane, select each of the 10 subsequent packets. Can you see any data/information in clear text, such as login name, devices information, and so on?

13. Scroll through the list of HTTPS packets and select the first packet that has an IPv6 source address of 2001:db8:1ab:1001::108.What is this packet number?
14. Looking in the Bytes pane, select each of the 10 subsequent packets. Can you see any data/information in clear text, such as login name, devices information, and so on?
15. What did you accomplish in this part of Capstone Lab 3? Summarize your answer.
16. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.

Question 2 – Please answer the question.

SSH

Question 4 – Please answer the question.

27218

Question 5 - Please answer the question.

Everything is encrypted, and I can see the initial handshake parameters in the first two packets

Question 6 - Please answer the question.

27364

Question 7 - Please answer the question.

The same as 3.4.5

Question 10 - Wireshark Filter.



Question 11 - Please answer the question.

27902

Question 12 - Please answer the question.

Everything is encrypted as expected since it is HTTPS. I can only see http 1.1 name

Question 13 - Please answer the question.

29910

Question 14 - Please answer the question.

The same as 3.4.12

Question 15 – Please answer lab summary question.

I learned how to visually analyze encrypted HTTPS traffic as it travels across a network and understand how the different layers of the OSI or TCP/IP model support its transmission.

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Capstone Lab 4

This capstone lab covers topics learned in Chapters 13 through 14 of *Guide to TCP/IP: IPv6 and IPv4*. It assumes you are working in a Windows 7 Professional or Windows 10 Pro environment. See the Preface for details about system requirements. To complete this lab, you must have Wireshark for Windows software installed and the trace (data) file named capstone-labs-4-trace-file.pcapng.

Capstone Lab 4.1

Capstone Lab 4.1: Create a Wireshark Display Filter to View IPv6 Messages

Time Required: 20 minutes

Objective: View screen capture outputs from a Windows 7 host, and use Wireshark to view a trace file and configure a display filter to view IPv6 messages from the multiple hosts.

Description: In this lab, you will view output from multiple commands run from a Windows 7 host (ipconfig command, an error message, and nslookup and ping commands) and use Wireshark to configure a specific display filter to view IPv6 messages. During the lab you will discover that, when the Windows 7 host executes the IPv6 Duplicate Address Detection (DAD) test, it effectively fails. In the ipconfig command output, the IPv6 address is listed as “duplicate” and an error message appears on the host. Then another IPv6 device is pinged on the network. An issue occurs that is caused by manual configuration of an IPv6 address on the Windows 7 host; the IPv6 address was already assigned via DHCPv6 to a Windows 10 host on the same network segment. When the Windows 7 host was reconfigured to support DHCPv6 and then received an appropriate Router Advertisement message, the Windows 7 host received a

valid/working IPv6 address. You need the capstone-labs-4-trace-file.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-4-trace-file.pcapng** file.
2. Configure a display filter to find all IPv6 packets
3. Apply the filter to the trace file.
4. In order to see the issue with the Windows 7 host, you will review a Windows 10 host DHCPv6 address assignment and its MAC address. Scroll down the packets list and select packet 142, which is the DHCPv6 server reply message to the host.
5. Expand **DHCPv6** in the packet details pane, expand **Identity Association for Non-temporary Address**, and expand **IA Address** to view the assigned IPv6 address. What is the assigned IPv6 address?
6. Go to and select packet 162 in the packet list pane. This is the Neighbor Solicitation message "to itself," better known as the DAD test.
7. Expand **Internet Control Message Protocol v6** in the packet details pane. What is the target address of the DAD test? What is the source MAC address?
8. Less than four minutes later, another DAD test message is seen on the network segment, this time from a different computer, but testing to the same IPv6 address as noted in Step 5. Scroll down the packets list and select packet 597. You should see two DAD test messages next to each other, one for the link-local address and one for the IPv6 Global Unicast Address (GUA) address.
9. If not already expanded, expand **Internet Control Message Protocol v6** in the packet details pane. What is the target address of the DAD test in packet 597? What is the source MAC address?
10. Go to and select packet 599 in the packet list pane. This is the Neighbor Advertisement reply message to the DAD test of packet 597.
11. If not already expanded, expand **Internet Control Message Protocol v6** in the packet details pane. What is the ICMPv6 Option (Target link-layer address)? Is this the same MAC address as recorded in Step 7 or Step 9?

12. Review the ipconfig command output shown in Figure CL4-1. What is the IPv6 GUA that is assigned to this Windows 7 host?

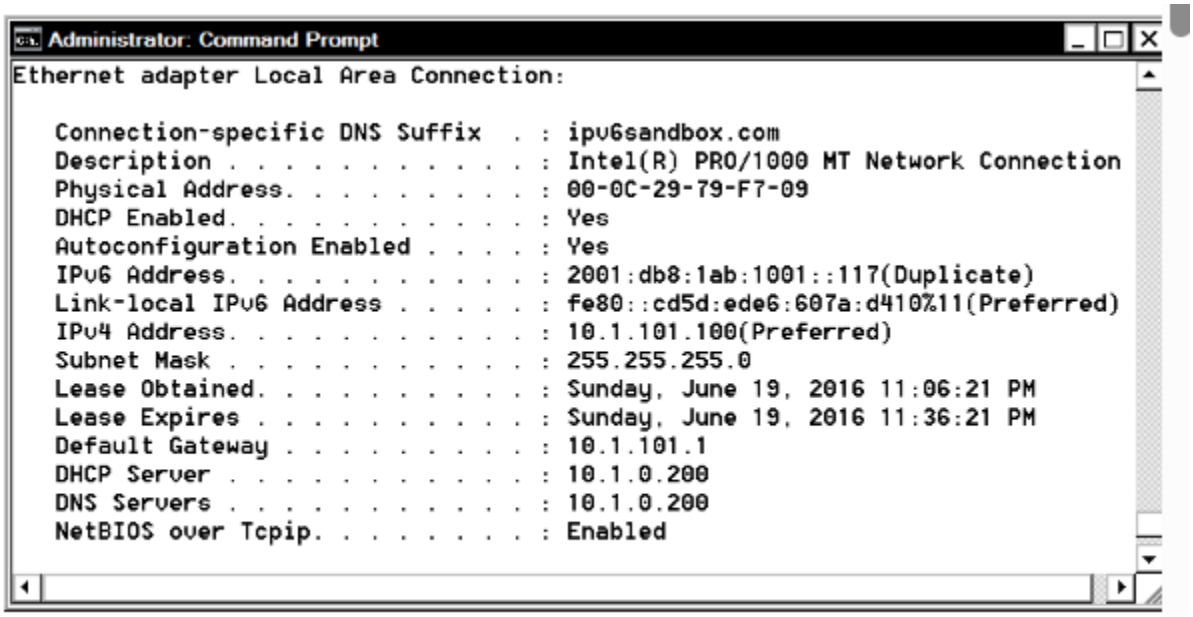


Figure CL4-1 Ipconfig command output

13. Review Figure CL4-2. This is an error message received on the Windows 7 host when it effectively failed the DAD test, because the Neighbor Advertisement reply message sent by the Windows 10 host indicates it is already assigned the IPv6 address (previous to the Windows 7 host wanting it).

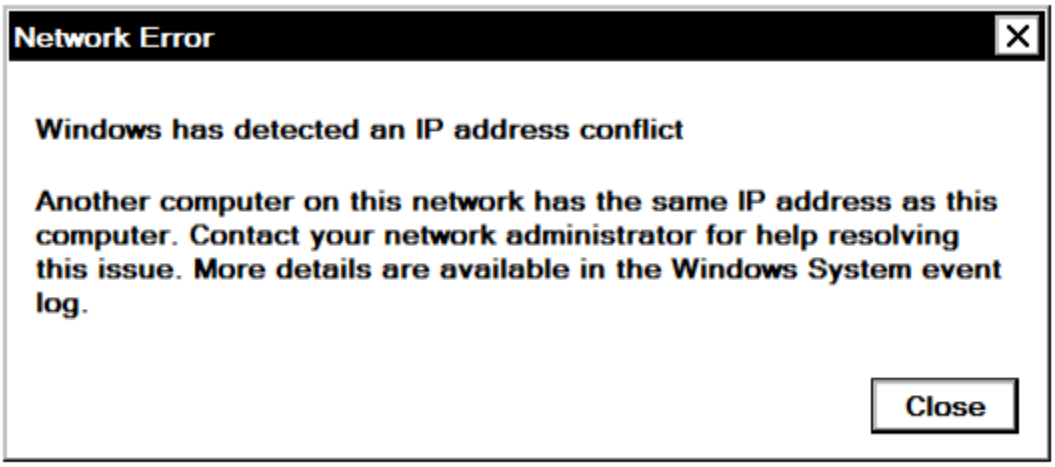


Figure CL4-2 Error message received on the Windows 7 host

14. Review Figure CL4-3. This figure shows that an nslookup command was executed on the Windows 7 host and the reply returned both IPv6 and IPv4 addresses for the requested Web site. Next, two ping commands were executed. Normally, when a Windows host (since Windows Vista) is dual-stacked and executes a ping, it will attempt to ping via IPv6 first. In the first ping, what IP address did the Windows 7 use to ping with? When a ping was executed and forced to use IPv6, what was the result?

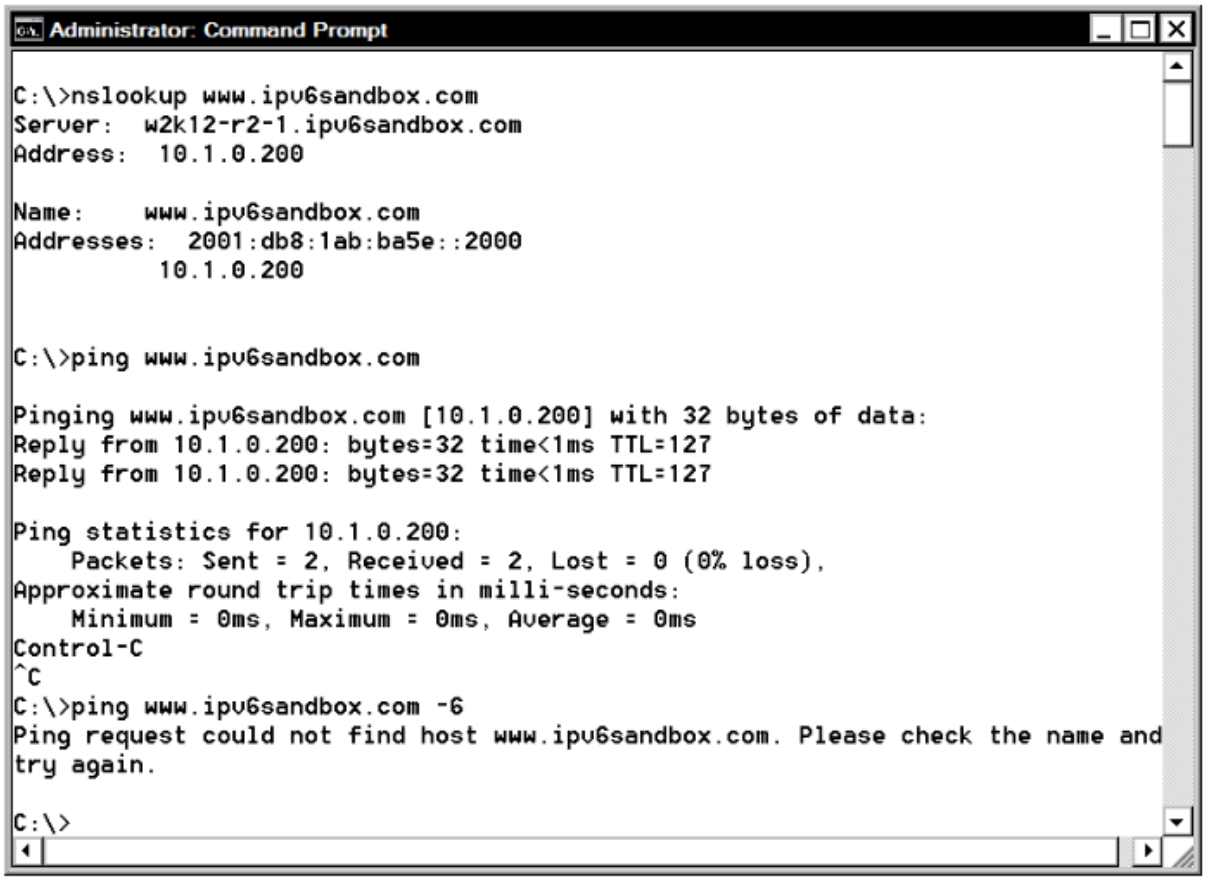


Figure CL4-3 Nslookup and ping commands executed on a Windows 7 host

15. What did you accomplish in this part of Capstone Lab 4? Summarize your answer.
16. Clear the display filter box.

Question 3 - Display Filter



Question 5 – Please answer the question.

IPv6 address: 2001:db8:1ab:1001::117

Question 7 – Please answer the question.

Target Address: 2001:db8:1ab:1001::117

MAC Address

Source: VMware_c8:71:8f (00:0c:29:c8:71:8f)

Question 9 – Please answer question.

Target Address: 2001:db8:1ab:1001::117

MAC Address

Source: VMware_79:f7:09 (00:0c:29:79:f7:09)

Question 11 - Please answer the question.

ICMPv6 Option (Target link-layer address : 00:0c:29:c8:71:8f)

The target link-layer layer address in Frame 599 is the same MAC Address in question #7

Question 12 - Please answer the question.

IPv6 Address. . . . . : 2001:db8:1ab:1001::117(Duplicate)

Question 14 - Please answer the question.

```
C:\>ping www.ipv6sandbox.com

Pinging www.ipv6sandbox.com [10.1.0.200] with 32 bytes of data:
Reply from 10.1.0.200: bytes=32 time<1ms TTL=127
Reply from 10.1.0.200: bytes=32 time<1ms TTL=127

Ping statistics for 10.1.0.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
```

The first ping used an IPv4 address, which is 10.1.0.200, but pinging to IPv6 failed.

Question 15 – Please answer lab summary question.

In this lab, I learned how to analyze IPv6 packets closely and identify target MAC addresses. I also learned that duplicate, which is DAD, IPv6 addresses cause address conflicts, preventing the affected host from using the duplicate address to communicate.

Capstone Lab 4.2

Capstone Lab 4.2: Create a Wireshark Display Filter to View ICMPv6 Router Advertisement and Ping Messages

Time Required: 20 minutes

Objective: View screen capture outputs from a Windows 10 host, and then use Wireshark to view a trace file and configure a display filter to view ICMPv6 Router Advertisement and ping command messages.

Description: In this lab, you will view output from multiple ipconfig and netsh commands from a Windows 10 host, and then use Wireshark to configure a specific display filter to view IPv6 messages. During the lab, you will discover that the Windows 10 host was specifically assigned its IPv6 address via DHCPv6 as seen in the ipconfig command output and starts a continuous ping to a Web server. Later, a different Router

Advertisement will be seen by this Windows 10 host, and it will configure additional IPv6 Stateless Address Autoconfiguration (SLAAC) addresses. The ping will continue but uses a different source address. A proper Router Advertisement will be seen and the SLAAC addresses will deprecate, and the ping continues. You need the capstone-labs-4-trace-file.pcapng file to complete the lab.

1. Review output from an ipconfig command in Figure CL4-4. What is the IPv6 Global Unicast Address (GUA) that is assigned to this Windows 10 host?

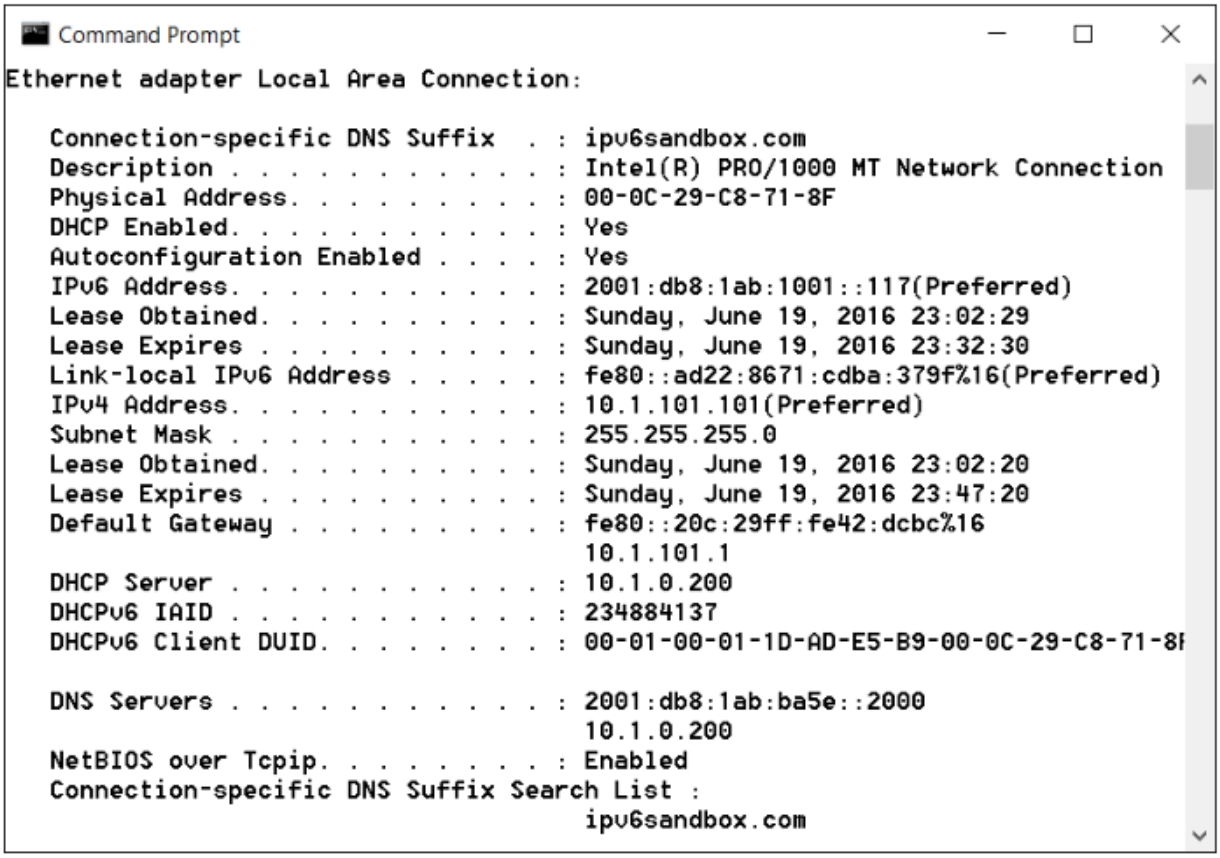


Figure CL4-4 Ipconfig command output

2. In Wireshark, open the **capstone-labs-4-trace-file.pcapng** file, if it's not already open.
3. Configure a display filter to find all IPv6 packets.
4. Apply the filter to the trace file.

5. With the IPv6 packets displayed, what packet is the first one where the Windows 10 host pings the Web server known as www.ipv6sandbox.com? (Hint: You will need to see DNS queries followed by the ping request. Begin looking around packet 6215.)
6. Scroll through the packets pane to the fifth Router Advertisement after the packet found in Step 5. What is this packet number?4
7. Select the Router Advertisement packet identified in Step 6 and observe its structure. In the packet details pane, expand **Internet Control Message Protocol v6** and also expand **Flags** to display the settings for the Managed address configuration and Other configuration options. From there, expand **ICMPv6 Option (Prefix information)** and then expand **Flag** to display the settings for the On-link flag(L) and Autonomous address-configuration flag(A) options.
8. While viewing this Router Advertisement packet, what are the settings of the M, O, L, and A flags?
9. Review the output from the ipconfig command shown in Figure CL4-5. What are the two additional IPv6 GUAs now assigned to this Windows 10 host? Bonus question: Which specific flag in the Router Advertisement caused the Windows 10 host to generate these two new addresses?

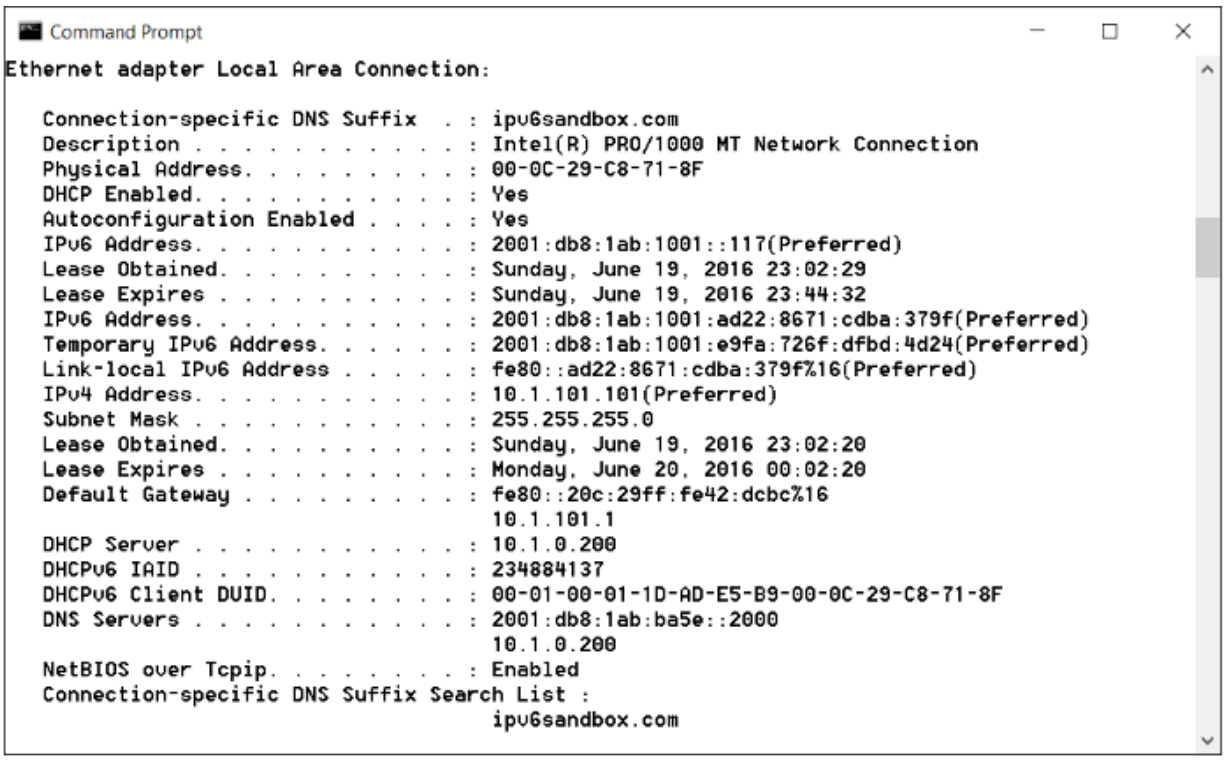


Figure CL4-5 Ipconfig command output

10. Scroll down the packets list pane to the next ping request packet to the Web server. What is this packet number? What is the source IPv6 address?
11. Again, review the output from the ipconfig command shown in Figure CL4-5. What is uniquely noted about the IPv6 address now being used for the ping requests? Bonus question: Why did the source IPv6 address change for the ping requests?
12. Scroll through the packets pane to the sixth Router Advertisement after the packet found in Step 6. What is this packet number?
13. Select the Router Advertisement packet identified in Step 12 and observe its structure. In the packet details pane, expand **Internet Control Message Protocol v6** and also expand **Flags** to display the settings for the Managed address configuration and Other configuration options. From there, expand **ICMPv6 Option (Prefix information)** and then expand **Flag** to display the settings for the On-link flag(L) and Autonomous address-configuration flag(A) options.

14. While viewing this Router Advertisement packet, what are the settings of the M, O, L, and A flags?
15. Review the output from the netsh command shown in Figure CL4-6. Focus on the Valid and Preferred Lifetime timer values for the SLAAC addresses.

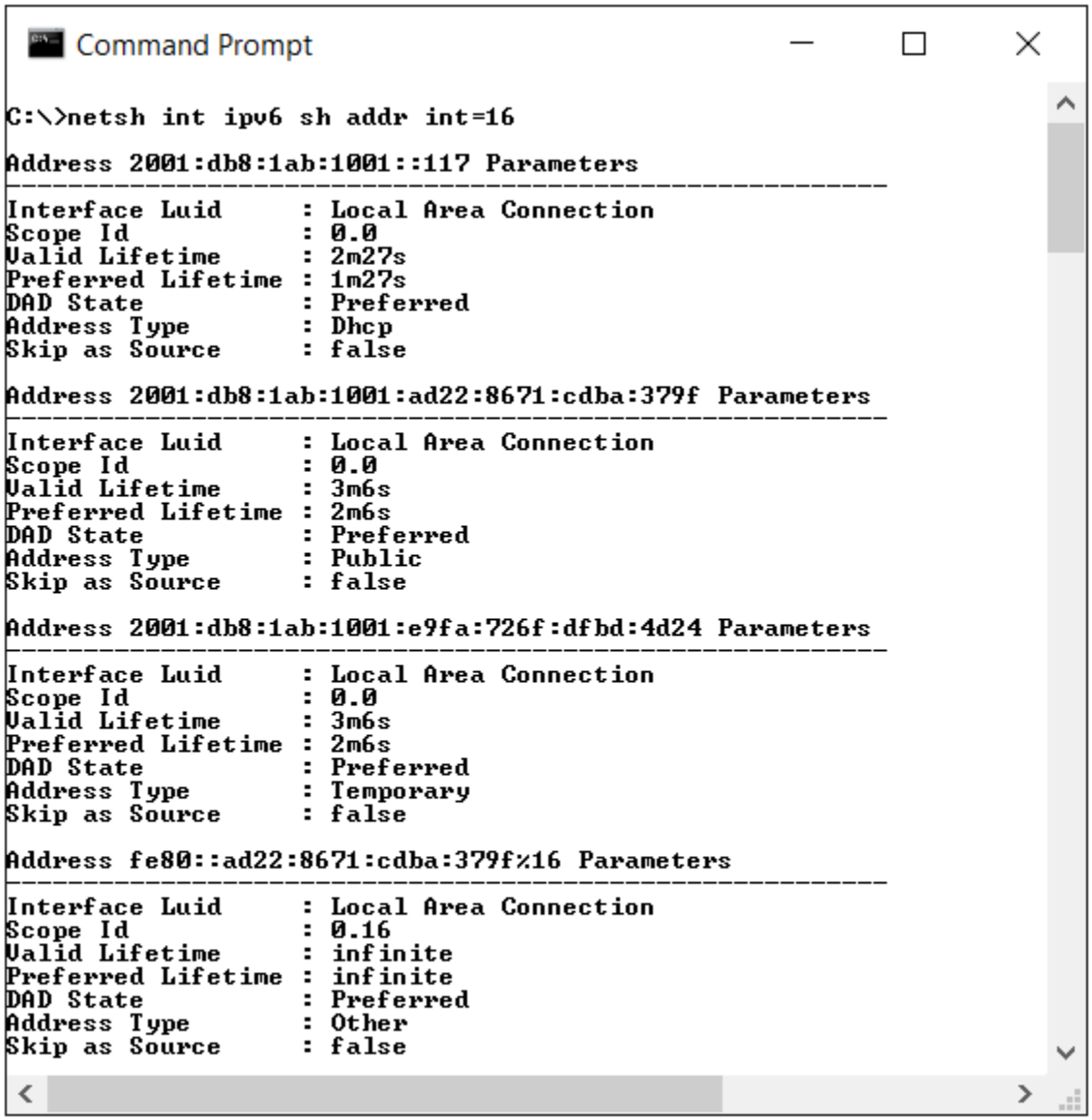


Figure CL4-6 Netsh command output

16. Review the output from the netsh command shown in Figure CL4-7, which is approximately one and half minutes after the output as seen in Step 15. Focus on

the Valid and Preferred Lifetime timer values for the SLAAC addresses. What differences do you see for SLAAC addresses compared to those values in Step 15?

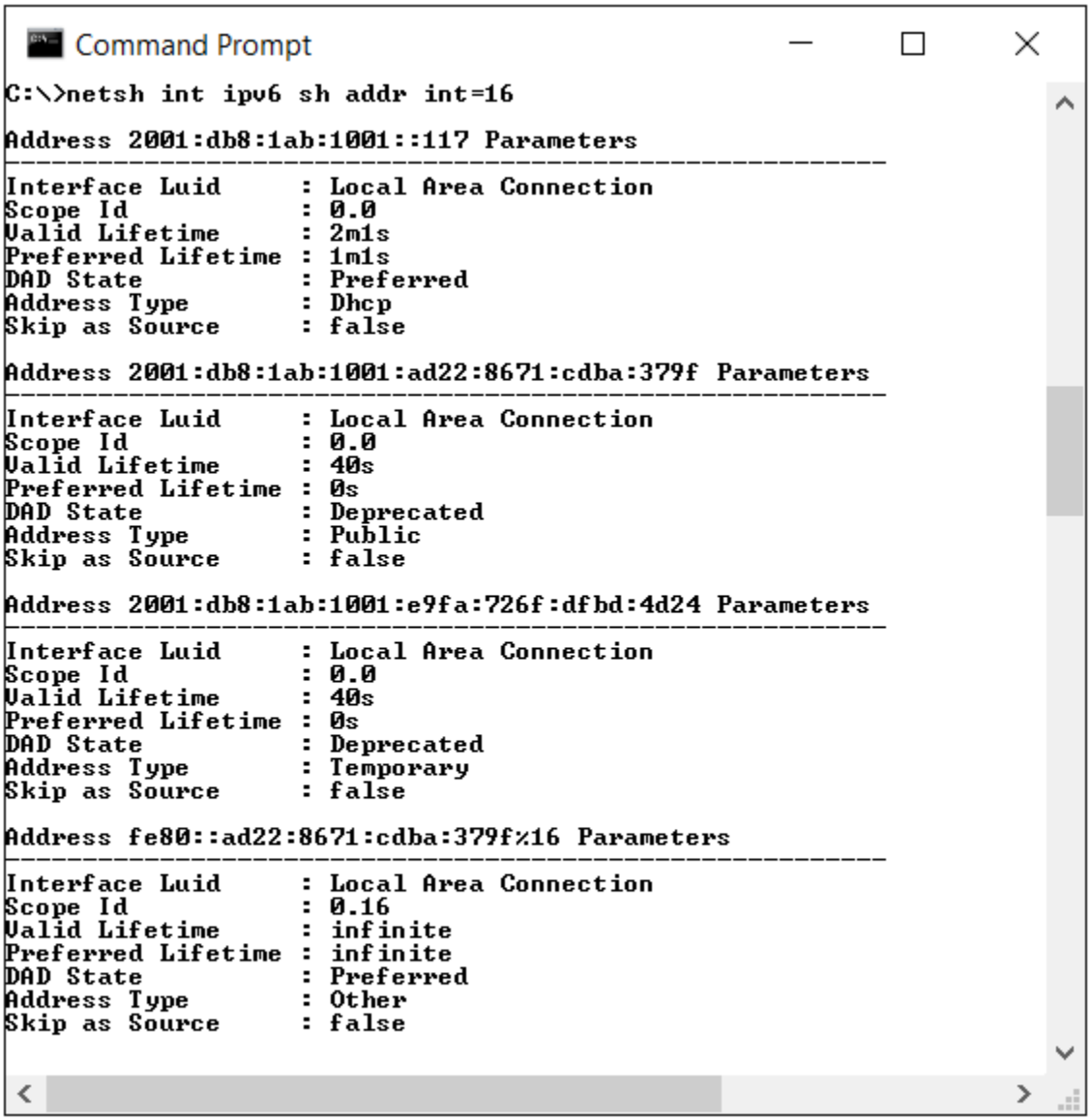


Figure CL4-7 Netsh command output

17. In Wireshark, scroll down the packets list pane to packet 8083. What is the source IPv6 address of this ping packet? Is it the same as the IPv6 address recorded in Step 1 or Step 10?
18. What did you accomplish in this part of Capstone Lab 4? Summarize your answer.

19. Clear the Wireshark display filter box.

Question 1 – Please answer the question.

```
IPv6 Address. . . . . . . . . . . : 2001:db8:1ab:1001::117(Preferred)
```

Question 3 - Wireshark Filter Display

1. (ipv6 && dns.qry.name == "www.ipv6sandbox.com") || icmpv6.type==134
2. ipv6

Question 5 - Please answer the question.

Frame 6221

Question 6 - Please answer the question.

Frame 6880

Question 8 - Please answer the question.

M = 1, O = 0, L = 1, A = 1

```
1... .... = Managed address configuration: Set
.0.. .... = Other configuration: Not set
1... .... = On-link Flag (L): Set
.1.. .... = Autonomous Address Configuration Flag (A): Set
```

Question 9 - Please answer the question.

```
IPv6 Address. . . . . . . . . . . : 2001:db8:1ab:1001:ad22:8671:cdba:379f(Preferred)
Temporary IPv6 Address. . . . . . : 2001:db8:1ab:1001:e9fa:726f:dfbd:4d24(Preferred)
```

BONUS: Flag A (Autonomous Address Configuration)

Question 10 - Please answer the question.

Frame 6788

Question 11 - Please answer the question.

What is uniquely noted about the IPv6 address now being used for the ping requests?

```
Temporary IPv6 Address. . . . . : 2001:db8:1ab:1001:e9fa:726f:dfbd:4d24(Preferred)
```

Bonus question: Why did the source IPv6 address change for the ping requests?

Because SLAAC created the new addresses, Windows preferred the temporary address for outgoing pings to improve privacy by avoiding consistent use of the same source address.

Wireshark: `Source Address: 2001:db8:1ab:1001::117`

Question 12 - Please answer the question.

Frame 7034

Question 14 - Please answer the question.

M = 1, O = 0, L = 1, A = 1

```
1... .... = Managed address configuration: Set
.0.. .... = Other configuration: Not set
```

```
1... .... = On-link Flag (L): Set
.1.. .... = Autonomous Address Configuration Flag (A): Set
```

Question 16 - Please answer the question.

For about 40 seconds

Question 17 - Please answer the question.

```
Source Address: 2001:db8:1ab:1001::117
```

It is the same IPv6 address recorded in Step 1

Question 18 – Please answer lab summary question.

I finally learned what SLAAC looks like in this lab. I also learned how RAs allow a computer to create IPv6 addresses and how it can switch the address used for ping requests. I learned that when an address's Preferred Lifetime reaches 0, it becomes deprecated and is no longer preferred for new connections.

Capstone Lab 4.3

Capstone Lab 4.3: Create a Wireshark Display Filter to View Active and Missing Router Advertisement Messages

Time Required: 20 minutes

Objective: View screen capture outputs from a Windows 10 host, and then use Wireshark to view a trace file and configure a display filter to view ICMPv6 Router Advertisement and ping messages.

Description: In this lab, you will view output from multiple commands (ipconfig and ping) run from a Windows 10 host, and use Wireshark to configure a specific display filter to view IPv6 messages. During the lab, you will discover that the Windows 10 host can successfully ping another IPv6 device on the network at one point but not later. The issue occurred because the router stopped sending its Router Advertisements. When the router started sending its Router Advertisements, the Windows 10 host was able to ping via IPv6 again. You need the capstone-labs-4-trace-file.pcapng file to complete the lab.

1. Review the output from the ipconfig command shown in Figure CL4-8. What is the IPv6 Global Unicast Address (GUA) that is assigned to this Windows 10 host? What is the default gateway address?

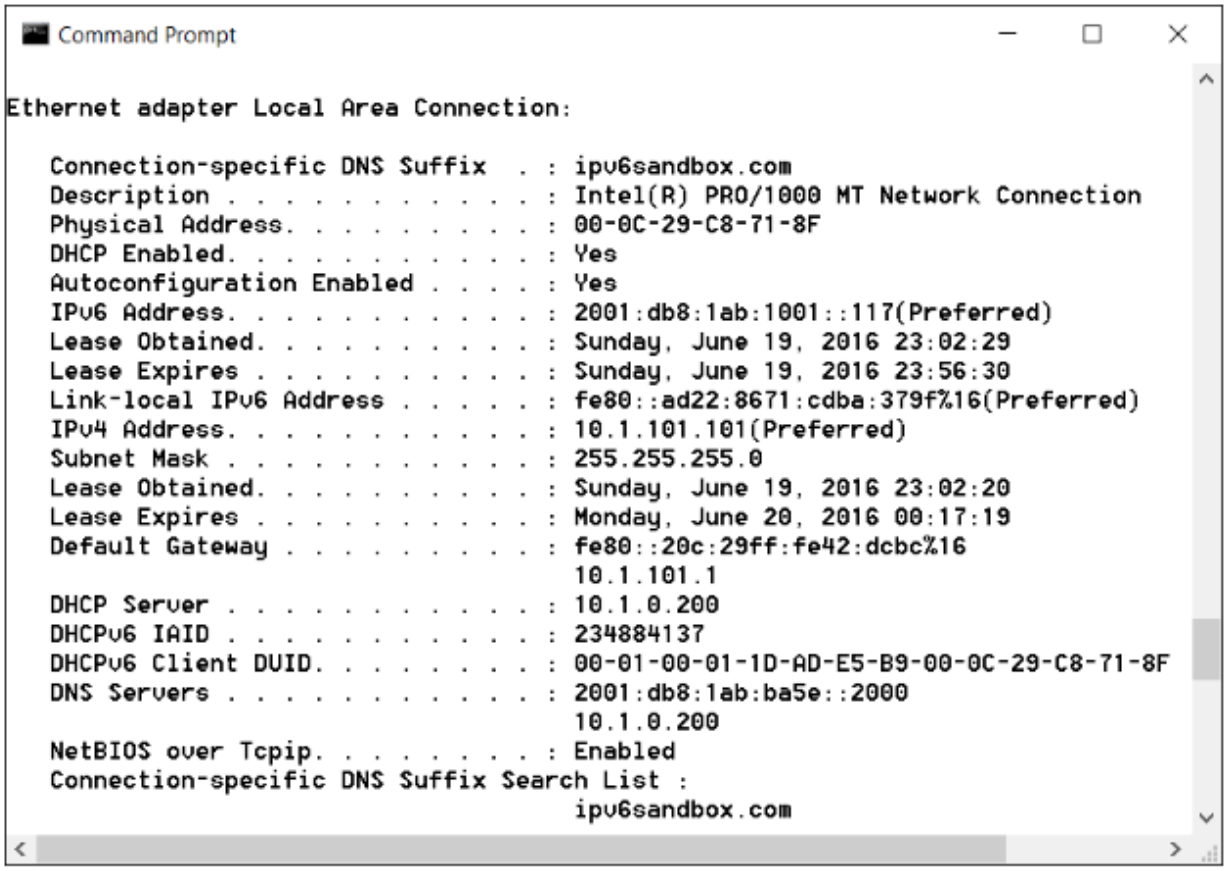


Figure CL4-8 Ipconfig command output

2. In Wireshark, open the **capstone-labs-4-trace-file.pcapng** file, if it's not already open.
3. Configure a display filter to find all IPv6 packets.
4. Apply the filter to the trace file.
5. Scroll down in the packets list pane and select packet 9633. This is considered a valid Router Advertisement.
6. In the packet details pane, expand **Internet Control Message Protocol v6** and also expand **Flags** to display the settings for the Managed address configuration and Other configuration options. From there, expand **ICMPv6 Option (Prefix information)** and then expand **Flag** to display the settings for the On-link flag(L) and Autonomous address-configuration flag(A) options.

7. While viewing this Router Advertisement packet, what are the settings of the M, O, L, and A flags? What is the router lifetime setting?
8. Scroll down the packets list approximately 20 or so packets and observe the host pinging the Web server. What is the packet number of the first ping request in this group of four successive pings?
9. Review the output from the ipconfig command shown in Figure CL4-9. What is different between this view and the information you recorded in Step 1?

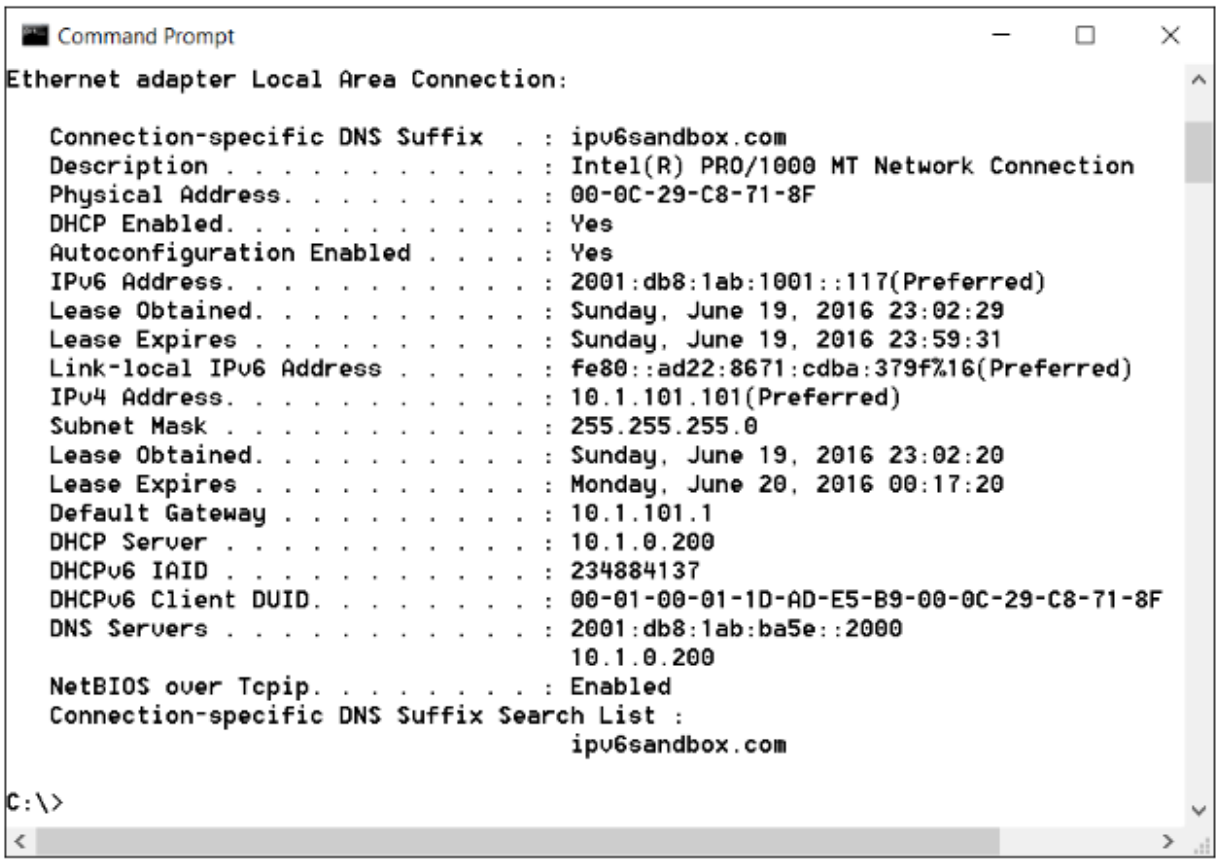


Figure CL4-9 Ipconfig command output

10. Review the screen capture shown in Figure CL4-10. A ping was executed to the Web server and it used IPv4. Then a second ping was executed to the Web server forcing it to use IPv6, which failed. Why did the IPv6 ping fail?

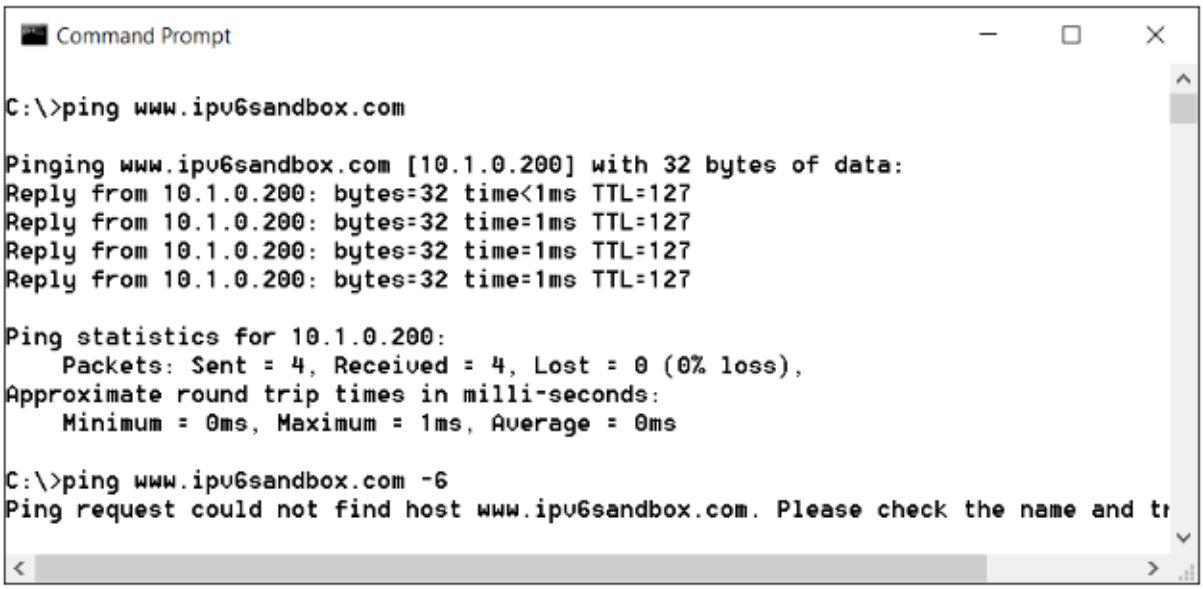


Figure CL4-10 Ping command output

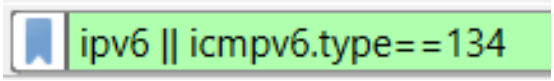
11. Scroll down the packets list and find the next two Router Advertisements. What are their packet numbers?
12. Select the first RA packet as recorded in Step 11. In the packet details pane, expand **Internet Control Message Protocol v6**. What is the router lifetime setting?
13. Select the second Router Advertisement packet as recorded in Step 11. In the packet details pane, expand **Internet Control Message Protocol v6**. What is the router lifetime setting?
14. What is the difference (if any) between the two router lifetime settings recorded in Steps 12 and 13?
15. Scroll down the packets list approximately 100 or so packets and observe the host pinging the Web server. What is the packet number of the first ping request in this group of four successive pings?
16. Bonus question: How did this host lose its IPv6 default gateway configuration while still having an IPv6 GUA?

17. What did you accomplish in this part of Capstone Lab 4? Summarize your answer.
18. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.

Question 1 – Please answer the question.

```
IPv6 Address. . . . . . . . . . . : 2001:db8:1ab:1001::117(Preferred)
Default Gateway . . . . . . . . . : fe80::20c:29ff:fe42:dcbc%16
```

Question 3 – Display Filter on Wireshark



Question 7 - Please answer the question.

M = 1, O = 0, L = 1, A = 0

```
1... .... = Managed address configuration: Set
.0.. .... = Other configuration: Not set
1... .... = On-link Flag (L): Set
.0.. .... = Autonomous Address Configuration Flag (A): Not set
```

Question 8 - Please answer the question.

Frame 9660

Question 9 - Please answer the question.

There are 2 default gateways in Figure CL4-8

Question 10 - Please answer the question.

It failed because the router stopped sending Router Advertisements, causing the computer's IPv6 default gateway to expire

Question 11 - Please answer the question.

Frame 9725

Frame 9887

Question 12 - Please answer the question.

Frame 9725 Router lifetime (s): 0

Frame 9887 Router lifetime (s): 270

Question 13 - Please answer the question.

Frame 9725 Preferred Lifetime: 180 (3 minutes)

Frame 9887 Preferred Lifetime: 180 (3 minutes)

Question 14 - Please answer the question.

Difference = 270 seconds in the 2nd RA Frame 9887

Question 15 - Please answer the question.

Frame 9991

(Optional Bonus Question) Question 16 - Please answer the question.

The host lost its IPv6 default gateway because the router advertised a Router Lifetime of 0, but kept its IPv6 GUA because the address's valid lifetime had not yet expired.

Question 17 (Required)– Please answer lab summary question.

I learned how to closely analyze Router Advertisements in a network with IPv6 autoconfiguration enabled and how Router Lifetime affects the host's default gateway. Although the steps were confusing to follow in the middle, the lab helped me understand why IPv6 communication can fail even when a host still has an IPv6 address